INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location	Microsoft, IA -- station KDSM, America/Chicago
Committed pair	OSM 830881733 -> 830881743 Microsoft / Microsoft
Facade gap	139.4 m between the two halls
Weather record	43,810 real hourly records from KDSM
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.2410 C at 225 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-11-10 (crossing)
Plant limit	18.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	gate off

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 12 of 24 hours with 1 mode change. That is 6 more than the reactive incumbent, at 0 unsafe hours against its 3. 4 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	12 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	6 of 24 hours free cooling, 2 mode change(s), 3 unsafe hour(s) 4 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	16.133	18.0	16.110	0.949
01	mechanical	yes	switch budget	16.413	18.0	16.670	0.840
02	mechanical	yes	switch budget	16.693	18.0	17.220	0.779
03	mechanical	yes	switch budget	17.744	18.0	18.890	0.698
04	mechanical	no	dry-bulb	18.086	18.0	18.890	0.895
05	mechanical	no	dry-bulb	18.358	18.0	18.895	0.792
06	mechanical	no	dry-bulb	19.208	18.0	18.895	0.648
07	mechanical	no	dry-bulb	20.045	18.0	18.890	1.035
08	mechanical	no	dry-bulb	21.422	18.0	19.445	1.359
09	mechanical	no	dry-bulb	22.538	18.0	19.446	1.668
10	mechanical	no	dry-bulb	23.762	18.0	20.565	1.812
11	mechanical	no	dry-bulb	18.363	18.0	9.440	1.806

12	FREE	yes	-	17.244	18.0	8.330	1.548
13	FREE	yes	-	16.710	18.0	7.220	1.379
14	FREE	yes	-	10.402	18.0	6.670	1.225
15	FREE	yes	-	9.090	18.0	6.110	1.107
16	FREE	yes	-	7.528	18.0	5.560	1.062
17	FREE	yes	-	6.146	18.0	4.440	1.012
18	FREE	yes	-	4.748	18.0	3.330	0.916
19	FREE	yes	-	3.924	18.0	2.220	1.240
20	FREE	yes	-	2.515	18.0	1.110	1.123
21	FREE	yes	-	1.382	18.0	-0.610	1.346
22	FREE	yes	-	0.282	18.0	-1.720	1.206
23	FREE	yes	-	-0.521	18.0	-2.220	1.040

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.133 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.412 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 16.693 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 17.744 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.086 C against a 18.0 C limit, so it fails by 0.086 C. It would take a limit 0.086 C higher, or a bound 0.086 C tighter, to change this hour.

05:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.358 C against a 18.0 C limit, so it fails by 0.358 C. It would take a limit 0.358 C higher, or a bound 0.358 C tighter, to change this hour.

06:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.208 C against a 18.0 C limit, so it fails by 1.208 C. It would take a limit 1.208 C higher, or a bound 1.208 C tighter, to change this hour.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 20.045 C against a 18.0 C limit, so it fails by 2.045 C. It would take a limit 2.045 C higher, or a bound 2.045 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.422 C against a 18.0 C limit, so it fails by 3.422 C. It would take a limit 3.422 C higher, or a bound 3.422 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.538 C against a 18.0 C limit, so it fails by 4.538 C. It would take a limit 4.538 C higher, or a bound 4.538 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.762 C against a 18.0 C limit, so it fails by 5.762 C. It would take a limit 5.762 C higher, or a bound 5.762 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.362 C against a 18.0 C limit, so it fails by 0.362 C. It would take a limit 0.362 C higher, or a bound 0.362 C tighter, to change this hour.

12:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.244 C, which is 0.756 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.548 C, and the margin is measured, not chosen: 1.518 C of group-conditional forecast error for this hour of day, plus 0.0295 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

13:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.710 C, which is 1.290 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.379 C, and the margin is measured, not chosen: 1.333 C of group-conditional forecast error for this hour of day, plus 0.0453 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

14:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.402 C, which is 7.598 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.225 C, and the margin is measured, not chosen: 1.184 C of group-conditional forecast error for this hour of day, plus 0.0418 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

15:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 9.090 C, which is 8.910 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.107 C, and the margin is measured, not chosen: 1.063 C of group-conditional forecast error for this hour of day, plus 0.0445 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

16:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 7.528 C, which is 10.472 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.062 C, and the margin is measured, not chosen: 1.034 C of group-conditional forecast error for this hour of day, plus 0.0279 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

17:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.146 C, which is 11.854 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.012 C, and the margin is measured, not chosen: 0.976 C of group-conditional forecast error for this hour of day, plus 0.0360 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

18:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 4.748 C, which is 13.252 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 0.916 C, and the margin is measured, not chosen: 0.888 C of group-conditional forecast error for this hour of day, plus 0.0279 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 3.924 C, which is 14.076 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.240 C, and the margin is measured, not chosen: 1.206 C of group-conditional forecast error for this hour of day, plus 0.0335 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 2.515 C, which is 15.485 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.123 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.0146 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 1.382 C, which is 16.618 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.346 C, and the margin is measured, not chosen: 1.324 C of group-conditional forecast error for this hour of day, plus 0.0216 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 0.282 C, which is 17.718 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.206 C, and the margin is measured, not chosen: 1.174 C of group-conditional forecast error for this hour of day, plus 0.0319 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is -0.521 C, which is 18.521 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.040 C, and the margin is measured, not chosen: 1.006 C of group-conditional forecast error for this hour of day, plus 0.0335 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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